

Curve fitting

Q1- Fit a straight-line to the data provided in the following table:

x_i	5	7	9	11	13
y_i	6	4.5	3.4	2.3	1.4

The equation of the line is:
 The value of coefficient of determination, r^2 is:

$$y = a_0 + a_1 x$$

$$a_1 = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

$$a_1 = \frac{5 \times 138.2 - 45 \times 17.8}{5 \times 445 - (45)^2} = -0.55$$

$$a_0 = \bar{y} - a_1 \bar{x}$$

$$a_0 = 3.56 - (-0.55)9 = 8.51$$

$$y = 8.51 - 0.55x$$

$$r^2 = \frac{S_t - S_r}{S_t}$$

$$S_t = \sum (y_i - \bar{y})^2$$

$$S_r = \sum (y_i - a_0 - a_1 x_i)^2$$

$$S_{t1} = (6 - 3.56)^2 = 5.95$$

$$S_{t2} = (4.5 - 3.56)^2 = 0.8836$$

$$\sum S_t =$$

$$S_{r1} = (6 - 8.51 + 0.55 \times 5)^2 = 0.0576$$

$$S_{r2} = (4.5 - 8.51 + 0.55 \times 7)^2 = 0.0256$$

$$\sum S_r =$$

Question 8 :

IF we fit a 2nd order polynomial for 8 values, the correlation coefficient is 0.9990, $\sum (y_i - \bar{y})^2 = 2500$
calculate :

1. The Standard error of estimate based on regression polynomial
2. Standard deviation.

$$\frac{S_y}{x} = \sqrt{\frac{S_r}{n-(m+1)}}$$

↪ Standard error

$$r = \sqrt{\frac{S_t - S_r}{S_t}}$$

$$0.999 = \sqrt{\frac{2500 - S_r}{2500}} \Rightarrow S_r = 4.9975$$

$$\frac{S_y}{x} = \sqrt{\frac{4.9975}{8-3}} = 0.997$$

$$S_y = \sqrt{\frac{\sum (y_i - \bar{y})^2}{n-1}} = \sqrt{\frac{2500}{8-1}} = 18.89$$

the following

x	1	2	3	4
y	1	200	400	900

data is regressed with least squares regression to
 $y = a_0 + a_1x$. The value of a_1 most nearly is

Answer: [redacted]

Answer: [redacted]

↳ linear

$$a_1 = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

Interpolation:

The following x, y data is given.

x	6	9	14
y	24	37	25

The Newton's divided difference second order polynomial for the data is given by $f_2(x) = b_0 + b_1(x-6) + b_2(x-6)(x-9)$. The value of b_1 is $+ b_2$.

Take $dx = 0.2$ in x -direction
3/8 rule in y -direction.
(10marks)

Answer:

3 points $\rightarrow 2^{\text{nd}}$ Order

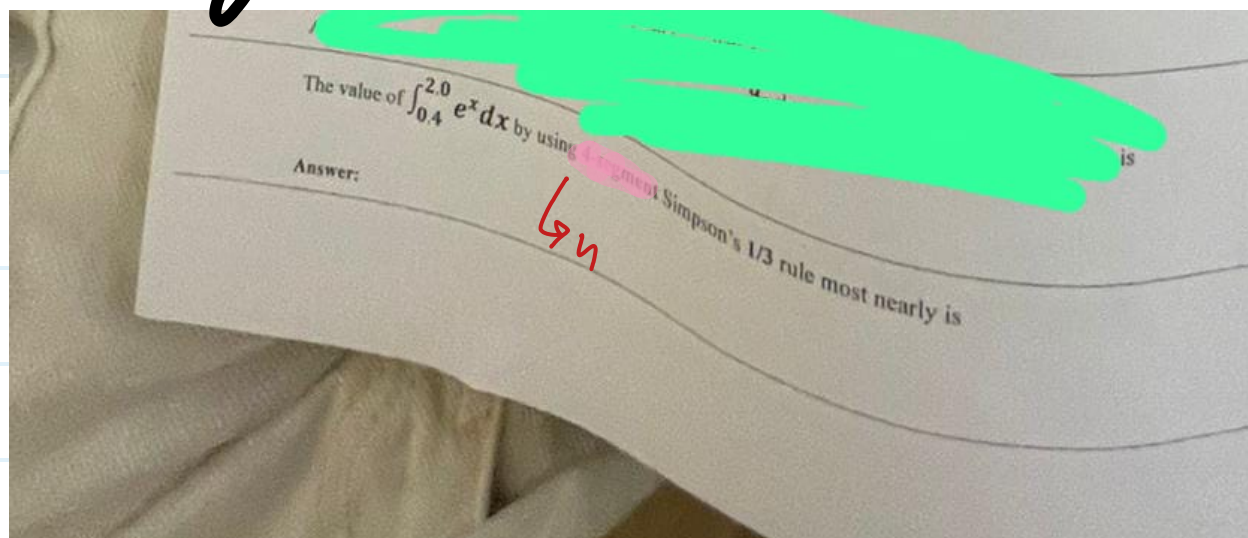
$$b_1 = \frac{\Delta y}{\Delta x} = \frac{37-24}{9-6} = 4.333 \quad b_{1(1)}$$

$$b_{1(2)} = \frac{25-37}{14-9} = -2.4$$

العدد الثاني (مثال)
العدد الأول (يتم)

$$b_2 = \frac{-2.4 - 4.333}{14-6} = -0.841$$

integration



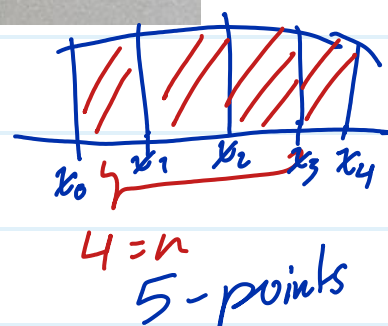
$$f(x) = e^x \quad \left. \begin{array}{l} n = 4 \\ 1 \quad x_0 = 0.4 \end{array} \right\}$$

$$\text{even} \leftarrow 2 \quad x_1 = 0.8$$

$$\text{odd} \leftarrow 3 \quad x_2 = 1.2$$

$$\text{even} \leftarrow 4 \quad x_3 = 1.6$$

$$\text{point} \leftarrow 5 \quad x_4 = 2$$



$$h = \frac{b-a}{n} = \frac{2-0.4}{4}$$

$$h = 0.4$$

Simpson 1/3 Rule:

$$\frac{h}{3} \left(f(x_0) + f(x_4) + 4 \left(\overset{\text{even}}{f(x_1) + f(x_3)} \right) + 2 \left(\overset{\text{odd}}{f(x_2)} \right) \right)$$

$$\frac{0.4}{3} (1.492 + 7.389 + 4(2.225 + 4.953) + 2(3.320)) = 5.898$$

Q3- Evaluate the following double integration: $\int_1^3 \int_1^3 (2x + y) dx dy$. (10 marks)

Take $dx = 0.5$ in the x-direction, $dy = 0.5$ in the y-direction. Apply the Trapezoidal rule in ~~the~~ x-direction and in the y-direction.

Answer:

$$f(x, y) = 2x + y$$

$$dx = h = 0.5$$

$$dy = k = 0.5$$

$\begin{matrix} x_i \\ y_j \end{matrix}$	1	1.5	2	2.5	3	
1	3	4	5	6	7	$\rightarrow I_1$
1.5	3.5	4.5	5.5	6.5	7.5	$\rightarrow I_2$
2	4	5	6	7	8	$\rightarrow I_3$
2.5	4.5	5.5	6.5	7.5	8.5	$\rightarrow I_4$
3	5	6	7	8	9	$\rightarrow I_5$

$\downarrow$
 I

x-direction: $I_1 = \frac{0.5}{2} (3 + 7 + 2 \times (4 + 5 + 6)) = 10$

$$I_2 = 11$$

$$I_3 = 12$$

$$I_4 = 13$$

$$I_5 = 14$$

y-direction: $I = \frac{k}{2} (I_1 + I_5 + 2 \times (I_2 + I_3 + I_4))$

$$= \frac{0.5}{2} (10 + 14 + 2 \times (11 + 12 + 13)) = 24$$

Q2- Compute the integration: $\int_2^4 \frac{1}{(x-1)^2} dx$, using Work to five significant figures

Apply Romberg integration taking $n=1$, $n=2$ & $n=4$ up to $R(3, 2)$.

Answer:

$$f(x) = \frac{1}{(x-1)^2}$$

(i, k)	1 st	2 nd
$R(1,0)$		
$R(2,0)$	$R(2,1)$	
$R(3,0)$	$R(3,1)$	$R(3,2)$

$$n=1 \quad \left| \quad h = \frac{b-a}{n} = \frac{4-2}{1} = 2 \right.$$

$$R(1,0) = \frac{h}{2} (f(2) + f(4)) = \frac{2}{2} \left(1 + \frac{1}{9} \right) = 1.1111$$

$$n=2 \quad \left| \quad h = \frac{4-2}{2} = 1 \right.$$

$$R(2,0) = \frac{1}{2} (f(2) + f(4) + 2 \times (f(3))) = 0.80550$$

$$n=4 \quad \left| \quad h = \frac{4-2}{4} = 0.5 \right.$$

$$R(3,0) = \frac{0.5}{2} (f(2) + f(4) + 2 \times (f(2.5) + f(3) + f(3.5))) = 0.69997$$

$$R(i, k) = \frac{4^k R(i, k-1) - R(i-1, k-1)}{4^k - 1}$$

$$R(2, 1) = \frac{4^{(1)} R(2, 0) - R(1, 0)}{4^{(1)} - 1} = \frac{4 \times 0.80550 - 1.111}{3} = 0.7036$$

$$R(3, 1) = \frac{4 R(3, 0) - R(2, 0)}{3} = 0.6647$$

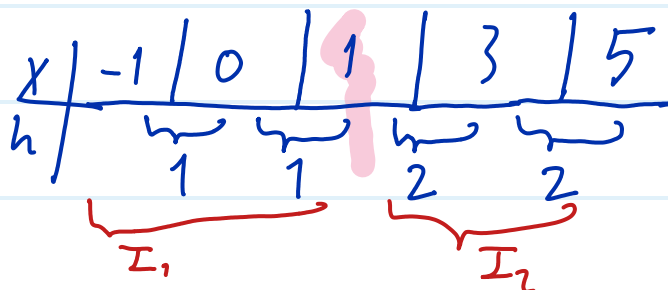
$$R(3, 2) = \frac{16 R(3, 1) - R(2, 1)}{15} = 0.6621$$

2-Find the area bounded by the curve and x axis from $x = -1$ to $x = 5$ using

Simpson's 1/3 rule

Answer:

	1	2	3	4	5
x :	-1	0	1	3	5
$f(x)$	3.1	0	4.3	7.5	2.2



$$n = 2 \quad h = 1$$

$$x_0 = -1$$

$$x_1 = 0$$

$$x_2 = 1$$

$$I_1 = \frac{1}{3} (f(-1) + f(1) + \underbrace{4 \times f(0)}_{\text{even}}) = 2.4666$$

$$n = 2 \quad h = 2$$

$$x_0 = 1$$

$$x_1 = 3$$

$$x_2 = 5$$

$$I_2 = \frac{2}{3} (f(1) + f(5) + \underbrace{4 \times f(3)}_{\text{even}}) = 24.333$$

$$I = I_1 + I_2 = 26.799$$

derivation

Question 1 :

Use **central** difference approximations to estimate the **second derivative** $O(h^2)$ of $y=e^{-x}\cos x$, at $x=\pi/4$, using a step size $h=\pi/12$.

Centered finite-divided difference formulas:

First Derivative

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h}$$

 $O(h^2)$

$$f''(x_i) = \frac{-f(x_{i+2}) + 8f(x_{i+1}) - 8f(x_{i-1}) + f(x_{i-2}))}{12h}$$

 $O(h^4)$

Second Derivative

$$f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))}{h^2}$$

 $O(h^2)$

$$f'''(x_i) = \frac{-f(x_{i+2}) + 16f(x_{i+1}) - 30f(x_i) + 16f(x_{i-1}) - f(x_{i-2}))}{12h^2}$$

 $O(h^4)$

Third Derivative

$$f'''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + 2f(x_{i-1}) - f(x_{i-2}))}{2h^3}$$

 $O(h^2)$

$$f^{(4)}(x_i) = \frac{-f(x_{i+3}) + 8f(x_{i+2}) - 13f(x_{i+1}) + 13f(x_{i-1}) - 8f(x_{i-2}) + f(x_{i-3}))}{8h^3}$$

 $O(h^4)$

$$f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))}{h^2}$$

$$f''\left(\frac{\pi}{4}\right) = \frac{f\left(\frac{\pi}{3}\right) - 2f\left(\frac{\pi}{4}\right) + f\left(\frac{\pi}{6}\right)}{\left(\frac{\pi}{12}\right)^2}$$

$$h = \frac{\pi}{12}$$

$$x_i = \frac{\pi}{4}$$

$$x_{i+1} = x_i + h$$

$$x_{i-1} = \frac{\pi}{3}$$

$$x_{i-1} = x_i - h$$

$$x_{i-1} = \frac{\pi}{6}$$

Q4: From the following data shown in the table below (15 marks)

$x:$	-1	0	1	3	5
$f(x)$	3.1	0	4.3	7.5	2.2

1- Find $f(2)$ - employing Newton's forward interpolation formula.

Answer:

5 point $\rightarrow$ 4th Order

x_i	y_i	b_1	b_2	b_3	b_4
-1	3.1	-3.1	3.7	-1.135	0.01
0	0	4.3	-0.9	-0.032	
1	4.3	1.6	-1.062		
3	7.5	-2.65			
5	2.2				

Forward

$$b_0 = 3.1$$

$$b_1 = -3.1$$

$$b_2 = 3.7$$

$$b_3 = -1.135$$

$$b_4 = 0.01$$

Backward

$$2.2$$

$$-2.65$$

$$-1.062$$

$$-0.032$$

$$0.01$$

$$f(x) = b_0 + b_1(x-x_0) + b_2(x-x_0)(x-x_1) + b_3(x-x_0)(x-x_1)(x-x_2) + b_4(x-x_0)(x-x_1)(x-x_2)(x-x_3)$$

$$f(x) = 3.1 + -3.1(x+1) + 3.7(x+1)x + -1.135(x+1)x(x-1) + 0.01(x+1)x(x-1)(x-5)$$

أفتران من الدرجة الرابعة

$$f(x) =$$

عوض 2 بعد التبسيط

$$f(2) =$$

Runge kutta

Answer:
Given $4\frac{dy}{dx} + 5y^2 = \sin x$, $y(0.2) = 9$
and using a step size of $h = 0.3$, the value of $y(0.8)$ using Euler's method is most nearly

$$4\frac{dy}{dx} + 5y^2 = \sin x$$

$$y(0.2) = 9$$

$$\frac{dy}{dx} = \frac{\sin x - 5y^2}{4}$$

$$y(0.5) = y(0.2) + \underbrace{f(0.2, 9)}_{\substack{\text{تعويض في} \\ \text{المشتقة}}} \cdot h = 9 + -101.2 \times 0.3$$

$$y(0.5) = -21.36$$

$$y(0.8) = y(0.5) + f(0.5, -21.36) \cdot h = -21.36 + -570.92 \times 0.3$$

$$y(0.8) = -192.64$$

Q5- Consider the ordinary differential equation: $\frac{dy}{dx} = x^2 - 2y^2 + x$

Given: $x_0 = 1, y_0 = 2, h = 0.5$

Use the Runge-Kutta method to the 4th order to find $y(1.5)$.

Answer:

$$\frac{dy}{dx} = x^2 - 2y^2 + x$$

$$y(1) = 2$$

$$\begin{aligned} K_1 &= h f(x_i, y_i) \\ &= 0.5 f(1, 2) = -3 \end{aligned}$$

$$\begin{aligned} K_2 &= h f\left(x_{i+\frac{h}{2}}, y_{i+\frac{K_1}{2}}\right) \\ &= 0.5 f(1.25, 0.5) = 1.15625 \end{aligned}$$

$$\begin{aligned} K_3 &= h f\left(x_{i+\frac{h}{2}}, y_{i+\frac{K_2}{2}}\right) \\ &= 0.5 f(1.25, 2.578) = -5.3239 \end{aligned}$$

$$\begin{aligned} K_4 &= h f(x_{i+h}, y_{i+K_3}) \\ &= 0.5 f(1.5, -3.3239) = -9.173 \end{aligned}$$

$$y_{i+1} = y_i + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$y(1.5) = 2 + \frac{1}{6} (-3 + 2(1.15625) + 2(-5.3239) + -9.173)$$

$$= -1.41805$$